Topic 3 — Discrete Probability Distributions

Random Variables

Definition

A **random variable** is a quantity which depends on the outcomes of a stochastic experiment. More rigorously, a random variable X is a function $X : \Omega \to \mathbb{R}$.

Example

Three fair six-sided dice are thrown simultaneously. Let X be the sum of scores on the dice. Then Ω consists of $\omega = (a, b, c)$ where $a, b, c \in \{1, \dots, 6\}$, and $X : \Omega \to \mathbb{R}$, $X(\omega) = a + b + c$. Thus, X may take only values from the finite set $\{3, 4, \dots, 18\}$, and hence, X is a discrete random variable.

Definition

If a random variable X takes values only from a discrete set $\{x_1, x_2, ...\}$, then X is called a **discrete random variable**. If X can take any values from an interval on the real line, then X is called a **continuous random variable**.

Probability Distributions and the CDF

Definition

A **probability distribution** of a random variable $X : \Omega \to \mathbb{R}$ is a mapping which assigns to each interval $E \subset \mathbb{R}$ the value of $\mathbb{P}(X \in E)$.

Definition

The **cumulative distribution function (CDF)** of a discrete random variable $X : \Omega \to \{x_1, x_2, ...\}$ is the function $F_X : \mathbb{R} \to [0, 1]$ defined by

$$F_X(x) = \sum_{x_i \leq x} \mathbb{P}(X = x_i) = \mathbb{P}(X \leq x).$$

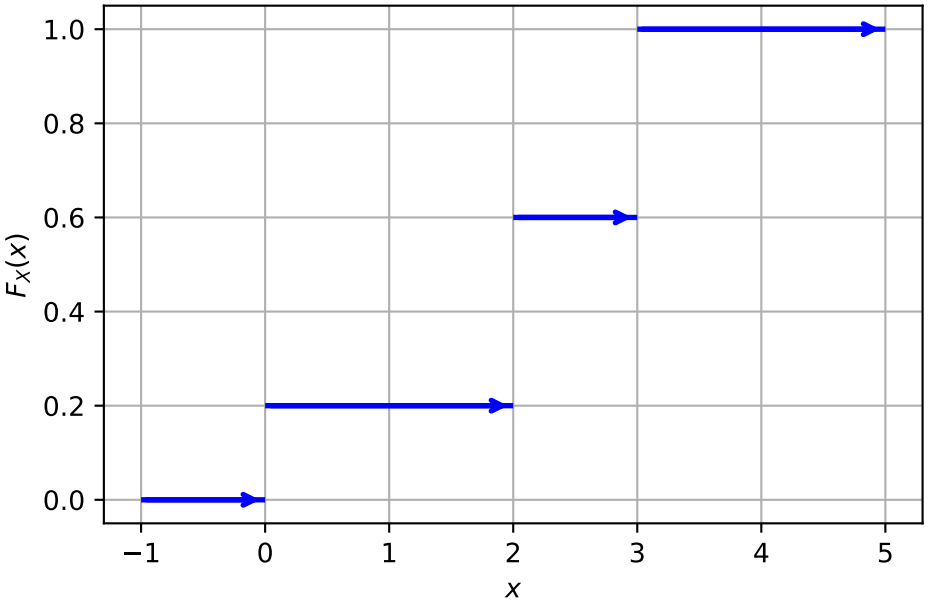


Figure 1: Cumulative function of the random variable $X : \Omega \to \{0, 2, 3\}$ with $\mathbb{P}(X = 0) = 0.2$, $\mathbb{P}(X = 2) = \mathbb{P}(X = 3) = 0.4$

Definition

A **discrete probability density function (discrete PDF)**, a.k.a. a **probability mass function (PMF)**, is a function that gives the probability that a discrete random variable is exactly equal to some value:

$$p_X(x) = \mathbb{P}(X = x).$$

Properties of F_X and p_X

Remark

- $0 \leq F_X(x) \leq 1$
- F_X is non-decreasing
- $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$
- $0 \leq p_X(x) \leq 1$
- If $X : \Omega \to \{x_1, x_2, ...\}$ then $p_X(x_1) + p_X(x_2) + ... = 1$

Signs matter here: for $X \in \mathbb{N} = \{1, 2, 3, ...\}$, $\mathbb{P}(5 \leq X < 7) \neq F_X(7) - F_X(5)$, instead

$$\mathbb{P}(5 \leq X < 7) = \mathbb{P}(4 < X \leq 6) \tag{1}$$
$$= F_X(6) - F_X(4). \tag{2}$$

Example: CDF of the Sum of Two Dice

Example

You throw two six-sided fair dice and calculate the sum of the numbers rolled. Let X be the random variable representing the sum. Find and sketch the CDF of X.
Solution: $\Omega = \{(a,b) \mid a,b \in \{1,2,\dots,6\}\}$, $X(\omega) = a+b \in \{2,\dots,12\}$, and

$$p_X(k) = \mathbb{P}(X=k) = \frac{\sharp\{(a,b) \mid a+b=k\}}{36}.$$

Example

$$\begin{aligned} p_X(2) &= \tfrac{1}{36}, & p_X(3) &= \tfrac{2}{36}, & p_X(4) &= \tfrac{3}{36}, & p_X(5) &= \tfrac{4}{36}, \\ p_X(6) &= \tfrac{5}{36}, & p_X(7) &= \tfrac{6}{36}, & p_X(8) &= \tfrac{5}{36}, & p_X(9) &= \tfrac{4}{36}, \\ p_X(10) &= \tfrac{3}{36}, & p_X(11) &= \tfrac{2}{36}, & p_X(12) &= \tfrac{1}{36}. \end{aligned}$$

$$F_X(x) = \begin{cases} 0, & x < 2, \\ \frac{1}{36}, & 2 \le x < 3, \\ \frac{3}{36}, & 3 \le x < 4, \\ \frac{6}{36}, & 4 \le x < 5, \\ \frac{10}{36}, & 5 \le x < 6, \\ \frac{15}{36}, & 6 \le x < 7, \\ \frac{21}{36}, & 7 \le x < 8, \\ \frac{26}{36}, & 8 \le x < 9, \\ \frac{30}{36}, & 9 \le x < 10, \\ \frac{33}{36}, & 10 \le x < 11, \\ \frac{34}{36}, & 11 \le x < 12, \\ 1, & x \ge 12. \end{cases}$$

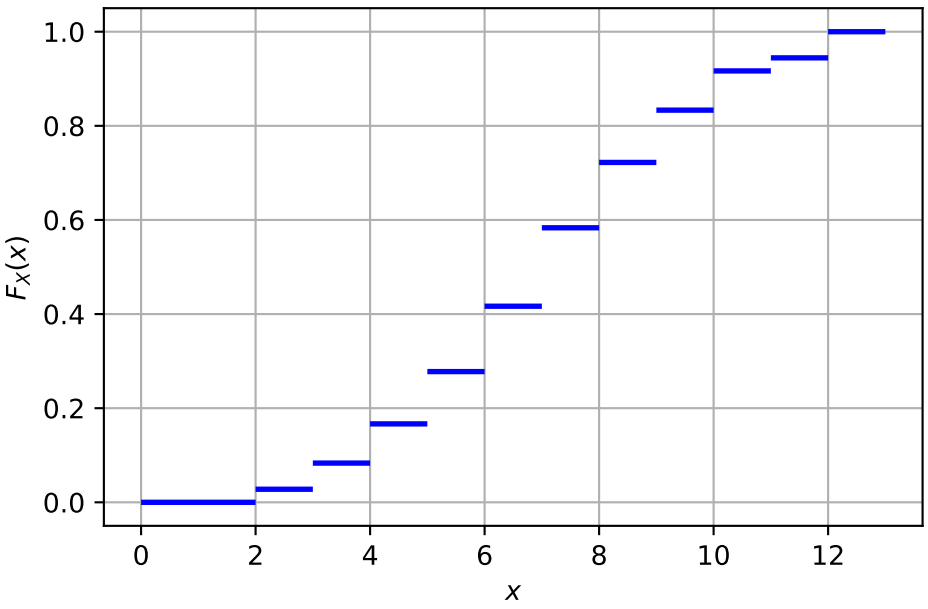


Figure 2: Cumulative distribution function of the sum of two fair dice

The Expected Value

Definition

The **expected value (mean)** $\mathbb{E}(X)$ of a random variable X is the average value it takes. If $X : \Omega \to \{x_1, x_2, ...\}$, then

$$\mathbb{E}(X) := \sum_i x_i \cdot p_X(x_i) = \sum_i x_i \cdot \mathbb{P}(X = x_i).$$

Example

For the sum of two dice above:

$$\mathbb{E}(X) = 2 \cdot \tfrac{1}{36} + 3 \cdot \tfrac{2}{36} + \cdots + 12 \cdot \tfrac{1}{36} = \frac{252}{36} = 7.$$

Example

Let $X \in \{100, -100\}$ with $\mathbb{P}(X = 100) = \mathbb{P}(X = -100) = \frac{1}{2}$. Then

$$\mathbb{E}(X) = 100 \cdot \tfrac{1}{2} + (-100) \cdot \tfrac{1}{2} = 0.$$

Remark

For any random variable $X : \Omega \to \mathbb{R}$ and any number $a \in \mathbb{R}$, $\mathbb{E}(aX) = a\mathbb{E}(X)$.
For any random variables $X, Y : \Omega \to \mathbb{R}$, $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$.

The Variance and Standard Deviation

Definition

The **variance** $\mathrm{Var}(X)$ of a random variable X is a measure of the spread of its values:

$$\mathrm{Var}(X) = \mathbb{E}\Big((X - \mathbb{E}(X))^2\Big) \tag{3}$$

$$= \mathbb{E}(X^2) - \big(\mathbb{E}(X)\big)^2 \geq 0. \tag{4}$$

Indeed:

$$\mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2 - 2\mathbb{E}(X) \cdot X + (\mathbb{E}(X))^2)$$

$$= \mathbb{E}(X^2) - 2\mathbb{E}(X) \cdot \mathbb{E}(X) + (\mathbb{E}(X))^2$$

(as the last term is not random)

$$= \mathbb{E}(X^2) - 2(\mathbb{E}(X))^2 + (\mathbb{E}(X))^2$$

$$= \mathbb{E}(X^2) - (\mathbb{E}(X))^2.$$

Example

For the sum of two dice: $\mathbb{E}(X^2) = \dfrac{1974}{36}$, and using $\mathbb{E}(X) = 7$,

$$\mathrm{Var}(X) = \frac{1974}{36} - 49 = \frac{35}{6} \approx 5.83.$$

Definition

The **standard deviation** of a random variable $X : \Omega \to \mathbb{R}$ is the square root of the variance of X: $\sigma(X) := \sqrt{\mathrm{Var}(X)}$.

Remark

For any random variable $X : \Omega \to \mathbb{R}$ and any number $a \in \mathbb{R}$,

$$\mathrm{Var}(aX) = a^2 \,\mathrm{Var}(X)$$

and

$$\sigma(aX) = |a|\sigma(X)$$

Example

Let $X \in \{-1, 1\}$ with $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = \frac{1}{2}$. Then $\mathbb{E}(X) = 0$, $\mathbb{E}(X^2) = 1$, so $\mathrm{Var}(X) = 1$ and $\sigma(X) = 1$.
Let $Y = 10X \in \{-10, 10\}$. Since $\mathrm{Var}(aX) = a^2 \,\mathrm{Var}(X)$,

$$\mathrm{Var}(Y) = 10^2 \cdot \mathrm{Var}(X) = 100, \qquad \sigma(Y) = \sqrt{100} = 10.$$

Independence of Random Variables

Definition

Two random variables $X : \Omega \to \mathbb{R}$ and $Y : \Omega \to \mathbb{R}$ are called **independent** if, for any $a, b \in \mathbb{R}$, the events $\{X \leq a\}$ and $\{Y \leq b\}$ are independent.

Remark

For any **independent** random variables $X, Y : \Omega \to \mathbb{R}$,

$$\mathbb{E}(XY) = \mathbb{E}(X) \cdot \mathbb{E}(Y), \tag{5}$$

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y). \tag{6}$$

Bernoulli Trials

Definition

Consider an experiment with only two possible outcomes: success (denoted by 1) and failure (denoted by 0). We will call such experiments **Bernoulli trials**.

Definition

A random variable X has the **Bernoulli distribution** if X can take only two values, usually 1 and 0. It models a Bernoulli trial. X is fully characterized by a single parameter $p \in [0, 1]$, the probability of success, i.e. its PMF is

$$p_X(1) = \mathbb{P}(X = 1) = p, \tag{7}$$
$$p_X(0) = \mathbb{P}(X = 0) = 1 - p. \tag{8}$$

Remark

Since $X = X^2$, we have that

$$\mathbb{E}(X) = \mathbb{E}(X^2) = 1 \cdot p + 0 \cdot (1 - p) = p, \tag{9}$$
$$\mathrm{Var}(X) = p - p^2 = p(1 - p). \tag{10}$$

The Binomial Distribution

Definition

The **binomial distribution** models the number of successes in a fixed number of independent and identically distributed Bernoulli trials. It is fully characterized by two parameters: n (the number of trials) and p (the probability of success in each trial). We denote this $X \sim Bin(n,p)$. The number k of successes in n trials may be any integer between 0 and n.

Remark

Recall that $n! = 1 \cdot 2 \cdot \ldots \cdot n$ is the **factorial** of n. We set $0! = 1$, and also $\binom{n}{0} = \binom{0}{0} = 1$.

Definition

The PMF of the Binomial distribution is:

$$p_X(k) = \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k},$$

where $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$.

Why? The probability of one specific arrangement with k successes then $n-k$ failures, e.g. $(\underbrace{1,\ldots,1}_{k},\underbrace{0,\ldots,0}_{n-k})$, is

$$\underbrace{p \cdots p}_{k} \cdot \underbrace{(1-p)\cdots(1-p)}_{n-k} = p^k (1-p)^{n-k},$$

and this is the same for *every* arrangement of k successes among n trials (e.g. $(\underbrace{1,\ldots,1}_{k-1},1,0,1,0\cdots 0)$ gives the same product). There are $\binom{n}{k}$ such arrangements, so we add up $\binom{n}{k}$ copies of $p^k(1-p)^{n-k}$.

Remark

We can write $X = Y_1 + \ldots + Y_n$ where $Y_1, \ldots, Y_n$ are independent random variables with identical Bernoulli distributions with parameter p. Then

$$\mathbb{E}(X) = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Remark

The sum of **all** values of the PMF is 1:

$$\sum_{k=0}^{n} \mathbb{P}(X=k) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = 1.$$

This is just the **binomial formula**.

Example: Winning Rounds (Binomial)

Example

In a game, a player has a 20% chance of winning each round. If the player plays 5 rounds, calculate the probability of winning exactly 3 rounds.
Solution: since $n = 5$ and $p = 0.2$,

$$\mathbb{P}(X = 3) = \binom{5}{3} (0.2)^3 (1 - 0.2)^{5-3} \tag{11}$$

$$= \frac{5 \cdot 4 \cdot 3}{1 \cdot 2 \cdot 3} \cdot 0.008 \cdot 0.64 \tag{12}$$

$$= 0.0512. \tag{13}$$

Example: Light Bulbs (Binomial)

Example

A company manufactures light bulbs, and 90% of them are of good quality, while the rest are defective. A customer buys 50 light bulbs.
1) What is the expected number of defective bulbs? Take "success" = "bulb is defective", so $p = 0.1$, $n = 50$:

$$\mathbb{E}(X) = 50 \cdot 0.1 = 5.$$

Remark

"Success" does not need to mean that something good happened — it depends on what we are calculating.

Example

2) Find the standard deviation of X:

$$\sigma(X) = \sqrt{\mathrm{Var}(X)} = \sqrt{np(1-p)} \tag{14}$$
$$= \sqrt{50 \cdot 0.1 \cdot 0.9} = \sqrt{4.5} \approx 2.12. \tag{15}$$

3) Find the probability that exactly 48 bulbs are defective:

$$\mathbb{P}(X = 48) = \binom{50}{48} \cdot 0.1^{48} \cdot 0.9^2 \tag{16}$$
$$= \binom{50}{2} \cdot 0.1^{48} \cdot 0.9^2 \tag{17}$$
$$= \frac{50 \cdot 49}{2} \cdot 10^{-48} \cdot 0.81. \tag{18}$$

4) Find the probability that at most 48 bulbs are defective:

$$\mathbb{P}(X \leq 48) = 1 - \mathbb{P}(X \geq 49) \tag{19}$$
$$= 1 - \binom{50}{49} 0.1^{49} \cdot 0.9 - \binom{50}{50} \cdot 0.1^{50} \tag{20}$$
$$= 1 - 50 \cdot 10^{-49} \cdot 0.9 - 10^{-50}. \tag{21}$$

Remark

Useful relation:

$$\binom{n}{k} = \binom{n}{n-k}.$$

For example, $\binom{50}{49} = \binom{50}{1} = 50$ and $\binom{50}{48} = \binom{50}{2} = \frac{50 \cdot 49}{2}$.

Example: Free Throws (Binomial Variance)

Example

A basketball player has a free throw success rate of 70%. If she attempts 20 free throws, find the variance of the number of successful free throws.
Solution: $p = 0.7$, $n = 20$:

$$\mathrm{Var}(X) = \sigma_X^2 = 20 \cdot 0.7 \cdot 0.3 = 4.2.$$

The Geometric Distribution

Definition

The **geometric distribution** models the number of trials needed to achieve the first success in a sequence of independent and identically distributed Bernoulli trials. It is characterized by a parameter p (the probability of success in each trial). We denote this $X \sim Geom(p)$. X can take any natural value $n = 1, 2, 3, ...$

Definition

We assume $p \neq 0$. The PMF of the geometric distribution is:

$$\mathbb{P}(X = n) = (1-p)^{n-1}p, \tag{22}$$

$$\mathbb{E}(X) = \frac{1}{p}, \tag{23}$$

$$\mathrm{Var}(X) = \frac{1-p}{p^2}. \tag{24}$$

Remark

$$\sum_{n=1}^{\infty} \mathbb{P}(X = n) = \sum_{n=1}^{\infty} (1-p)^{n-1}p \tag{25}$$

$$= \frac{p}{1-(1-p)} = 1. \tag{26}$$

Example: Multiple-Choice Guessing (Geometric)

Example

A student is preparing for a multiple-choice exam, where each question has 4 choices, and only one is correct. If the student guesses the answers, what is the probability that the first correct answer occurs on the third guess? How many guesses would the student need on average?
Solution: $p = \frac{1}{4}$,

$$\mathbb{P}(X = 3) = \left(1 - \tfrac{1}{4}\right)^2 \cdot \tfrac{1}{4} \tag{27}$$

$$= \frac{9}{64}, \qquad \mathbb{E}(X) = \frac{1}{p} = 4. \tag{28}$$

The Negative Binomial Distribution

Definition

The **negative binomial distribution** models the number of failures in a sequence of independent and identically distributed Bernoulli trials before a specified number of successes occurs. It is characterized by two parameters: r (the number of successes) and p (the probability of success in each trial). We denote this $X \sim NB(r, p)$. If $X = k$ is the number of failures, the total number of trials is $n = k + r$.

Definition

The PMF of the negative binomial distribution is:

$$\mathbb{P}(X = k) = \binom{k + r - 1}{k} p^r (1 - p)^k, \tag{29}$$

$$\mathbb{E}(X) = \frac{r(1 - p)}{p} = \frac{r}{p} - r, \tag{30}$$

$$\text{Var}(X) = \frac{r(1 - p)}{p^2}. \tag{31}$$

Example: Free Throws (Negative Binomial)

Example

A student is practicing basketball free throws with success probability 0.7. The student stops as soon as they achieve 3 successful free throws. What is the probability that by that time the student will have had 2 failures?

Remark

Equivalent formulation: what is the probability that it will take the student exactly 5 trials to make 3 successful free throws?

Example

Solution: $p = 0.7$, $r = 3$, $k = 2$:

$$\mathbb{P}(X = 2) = \binom{2+3-1}{2} \cdot (0.7)^3 \cdot (0.3)^2 \tag{32}$$

$$= \frac{4 \cdot 3}{2} \cdot 0.49 \cdot 0.09 = 0.2646. \tag{33}$$

Example: Quality Control (Negative Binomial)

Example

In a quality control process, a manufacturer wants to inspect several items to find 2 defective items. If the probability of finding a defective item is 0.1, what is the expected number of items that need to be inspected?

Solution: $p = 0.1$, $r = 2$. The number X of items to inspect is $X = Y + 2$, where $Y \sim NB(2, 0.1)$ counts the "failures" (non-defective items found along the way). Then

$$\mathbb{E}(Y) = \frac{2}{0.1} - 2 = 18, \qquad \mathbb{E}(X) = \mathbb{E}(Y) + 2 = 20.$$

The Poisson Distribution

Definition

The **Poisson distribution** models the number of independent events occurring in a fixed interval of time or space. It is characterized by a single parameter $\lambda > 0$ (the average rate of events per interval *of the same size*). We denote this by $X \sim Po(\lambda)$.

Definition

The PMF of the Poisson distribution is:

$$\mathbb{P}(X = n) = \frac{\lambda^n}{n!} e^{-\lambda}.$$

The expected value and the variance of the Poisson random variable are equal:

$$\mathbb{E}(X) = \mathrm{Var}(X) = \lambda.$$

Remark

$$\sum_{n=0}^{\infty} \frac{\lambda^n}{n!} e^{-\lambda} = 1.$$

Remark

It is crucial to use, for λ, the average rate of events per interval **under investigation** — it should be recalculated to match the interval considered.

Example: Call Centre (Poisson)

Example

In a call center, calls arrive at an average rate of 5 calls per minute. Calculate the probability that
a) exactly 15 calls will arrive in the next 2 minutes;
b) at least 2 calls will arrive in the next 30 seconds.

Example

a) Rate per 2 minutes is $2 \cdot 5 = 10$, so $\lambda = 10$, $X \sim Po(10)$:

$$\mathbb{P}(X = 15) = \frac{10^{15}}{15!} e^{-10} \approx 0.0347.$$

Example

b) Rate per 30 seconds (0.5 min) is $0.5 \cdot 5 = 2.5$, so $\lambda = 2.5$, $X \sim Po(2.5)$. Using the complement,

$$\begin{aligned} \mathbb{P}(X \leq 1) &= \frac{2.5^0}{0!} e^{-2.5} + \frac{2.5^1}{1!} e^{-2.5} && (34) \\ &= 3.5 e^{-2.5} \approx 0.2873, && (35) \end{aligned}$$

$$\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X \leq 1) \approx 0.7127.$$

Remark

If λ is the average rate *per unit time*, then the PMF of the distribution of events occurring in a time interval of length t is

$$\mathbb{P}(X = n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}.$$